

Unified Acoustic Map-Based Instance Modeling for Semantic Acoustic Imaging and Sound Event Localization

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Abstract—Semantic acoustic imaging (SAI) has recently attracted significant attention because it provides dense spatial representations of sound sources, offering richer localization information than conventional sparse point-wise direction estimates. However, estimating high-resolution semantic acoustic maps from low-resolution sensor observations remains highly challenging. To address this problem, we present a unified acoustic-map-based instance modeling framework for DCASE 2026 Task 3 that supports both the audio-only and audio-visual tracks. The proposed system transforms four-channel microphone signals into UpLAM-derived equirectangular acoustic maps, which serve as dense spatial evidence on a unified azimuth-elevation representation. For the audio-only track, the acoustic maps are processed by a Mask R-CNN-inspired instance detection framework featuring a ResNet-FPN backbone, RoI-level energy-mask prediction, full-frame energy supervision, auxiliary distance regression, and temporal filtering. For the audio-visual track, the same instance-prediction framework is extended with modality-specific feature extractors, acoustic-guided feature-pyramid fusion, a ResNet-Conformer context adapter, and RoI-level cross-modal attention. In addition, teacher-guided semantic priors and region-aware energy export are introduced to improve category assignment, source-count estimation, and mask compactness. Experimental results on the official evaluation set demonstrate the effectiveness of the proposed framework, achieving third place in Track A and first place in Track B. The first-ranked Track B submission achieved a Macro mAP of 0.0131 and a Macro Pearson correlation of 0.4002, while a Pearson-oriented variant reached a higher Macro Pearson correlation of 0.5634.

Index Terms—Semantic acoustic imaging, sound event localization and detection, UpLAM, Mask R-CNN, audio-visual fusion

1. INTRODUCTION

Spatial perception is a fundamental capability that enables autonomous systems, such as service robots, intelligent vehicle cockpits, and security monitoring platforms, to interpret complex real-world environments. Sound event localization and detection (SELD) addresses this need by jointly estimating the temporal activity, semantic category, and spatial position of active sound events [1], [2]. Most existing SELD frameworks adopt a point-source assumption, in which each sound event is represented by a direction-of-arrival (DoA) vector consisting of azimuth and elevation angles, optionally supplemented with a distance estimate [3], [4]. Although this representation is compact and effective for many spatial audio tasks, it is inherently limited when describing sound sources whose acoustic energy extends over a spatial region [5]. In practical environments, the emitted acoustic energy of real-world sources is often spatially distributed rather than concentrated at a single point. Conventional SELD formulations are therefore insufficient for accurately characterizing complex spatial energy distributions. Recent learning-based SELD studies have further investigated environment generalization and large-scale pre-training, highlighting the need for robust spatial audio representations [6], [7].

To overcome these representational limitations, DCASE 2026 Task 3 introduces SAI for SELD, in which each active event is represented

as a class-labeled acoustic energy region on a sphere [8]. The target masks are defined in an equirectangular coordinate system. The task includes an audio-only track, which uses four-channel spatial audio, and an audio-visual track, which additionally incorporates synchronized panoramic video. Both tracks are highly challenging because the systems must predict dense, high-resolution SAI masks from input observations with limited spatial resolution. In this sense, the task can be viewed as a spatial acoustic super-resolution problem: during evaluation, detailed acoustic images must be reconstructed from four-channel observations that are much sparser than the high-channel reference annotations [8]. In the audio-visual setting, visual information can provide complementary cues for object identity and spatial extent. However, simple early fusion of RGB frames and acoustic maps may fail to fully exploit their complementary cues; recent AV-SELD studies therefore emphasize structured fusion, teacher-student learning, semantic representations, and audio-guided visual attention [9]–[12].

In this work, we address both tracks using a unified instance-level SAI framework and develop dedicated systems for the audio-only and audio-visual settings. The core representation is an UpLAM-derived acoustic map [13], which is projected onto the same equirectangular coordinate system as the target SAI masks. Based on this representation, a Mask R-CNN-style detector [14] predicts sound event instances, category scores, spatial regions, energy masks, and distances. For the audio-only track, the framework operates directly on acoustic maps. For the audio-visual track, we extend the framework with acoustic-guided fusion at the feature-pyramid level and cross-modal fusion at the RoI level. The main contributions of this work are summarized as follows:

- We formulate both Track A and Track B of DCASE 2026 Task 3 within a unified acoustic map-based instance-prediction framework for semantic acoustic imaging.
- We improve audio-only SAI inference through local energy-mask prediction, global energy supervision, distance regression, balanced sampling, and temporally constrained post-processing.
- We develop an audio-visual extension that integrates modality-specific feature pyramids, gated acoustic guidance, a ResNet-Conformer context adapter, RoI-level cross-modal attention, and inference-time semantic priors.

2. PROPOSED METHOD

2.1. Acoustic map representation

The audio front-end converts each 100-ms block of four-channel microphone signals into a spatial acoustic representation. Following the UpLAM-based baseline pipeline [13], [15], multichannel time-frequency features are first computed from the four-channel signals and then compressed into nine frequency bands between 1.5 and 4.5 kHz. This frequency-band compression is used for two reasons. First, it reduces spectral redundancy and keeps the acoustic representation

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compact enough for the subsequent instance detector. Second, the selected mid-frequency range preserves spatially informative inter-channel cues that are useful for UpLAM-based acoustic localization, while avoiding excessive sensitivity to very low-frequency diffuse energy and high-frequency noise. A pretrained UpLAM model estimates band-wise spatial activations, which are projected onto a 180×360 equirectangular projection (ERP) grid. The resulting tensor $\mathbf{A} \in \mathbb{R}^{9 \times 180 \times 360}$ retains frequency-dependent spatial energy patterns and serves as the basic spatial evidence for Track A.

For Track B, the same acoustic ERP map is aligned with the panoramic RGB frame by matching the azimuth direction of the acoustic image to the visual coordinate system. This shared canvas is important because the final prediction is not a point coordinate but a class-specific energy region. During training, we apply cyclic azimuth shifts, band masking, and mild Gaussian perturbations to the acoustic maps, where band masking follows the idea of frequency masking in SpecAugment [16]. In Track B, horizontal flips are synchronized across audio maps, video frames, and labels; visual color jittering is applied only to the RGB stream.

2.2. Shared instance prediction framework

The shared detector follows the Mask R-CNN paradigm with a ResNet-50-FPN backbone [14], [17], [18]. Given an acoustic map or fused audio-visual feature pyramid, the region proposal network generates candidate event regions on the ERP canvas. RoIAlign extracts local features from each proposal, after which parallel heads estimate the event class, confidence score, bounding box, RoI energy mask, and source distance. This instance-based design avoids treating the whole spherical image as a single dense regression target and is better suited for scenes containing multiple simultaneous sound events.

Training jointly optimizes instance detection and acoustic-energy reconstruction. With the standard binary mask branch replaced by a continuous energy-mask head, the detection loss is

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{obj}} + \mathcal{L}_{\text{rpn-box}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{roi-box}}. \quad (1)$$

Here, $\mathcal{L}_{\text{obj}}$ is binary cross-entropy, both box terms use Smooth L1 loss with $\beta = 1/9$ [19], and $\mathcal{L}_{\text{cls}}$ is label-smoothed, class-weighted cross-entropy with inverse-square-root class weights and centered log-prior adjustment.

For each sampled RoI, the matched continuous energy field and binary support mask are resized to 28×28 . Let $\hat{\mathbf{E}}$, $\mathbf{E}$, and $\mathbf{M}$ denote the prediction, continuous target, and support mask, and let $\text{MSE}_{\mathbf{M}}$ denote MSE over pixels with $\mathbf{M} = 1$. The RoI energy loss is

$$\begin{aligned} \mathcal{L}_{\text{r}} = \lambda_{\text{r,E}} \left[\text{MSE}(\hat{\mathbf{E}}, \mathbf{E}) + \alpha_{\text{r}} \text{MSE}_{\mathbf{M}}(\hat{\mathbf{E}}, \mathbf{E}) \right] \\ + \lambda_{\text{r,I}} \mathcal{L}_{\text{IoU}}(\hat{\mathbf{E}}, \mathbf{M}) + \lambda_{\text{r,L}} \mathcal{L}_{\text{Lov}}(\hat{\mathbf{E}}, \mathbf{M}). \end{aligned} \quad (2)$$

The MSE terms reconstruct energy values, while soft-IoU and Lovasz-Jaccard [20] improve overlap and compactness.

A full-frame decoder provides scene-level supervision. For frame b , $\mathbf{G}_b$ is the pixel-wise maximum over instance maps, $\mathbf{U}_b$ their support union, and $w_{b,p} = \alpha_g$ inside $\mathbf{U}_b$ and 1 otherwise. The global reconstruction loss is

$$\mathcal{L}_{\text{g}} = \lambda_{\text{g,E}} \frac{\sum_{b,p} w_{b,p} (\hat{G}_{b,p} - G_{b,p})^2}{\sum_{b,p} w_{b,p}} + \lambda_{\text{g,L}} \mathcal{L}_{\text{Lov}}(\hat{\mathbf{G}}, \mathbf{U}), \quad (3)$$

where $\hat{\mathbf{G}}$ is the predicted full-frame energy map; this term preserves scene-level coverage while the RoI loss refines individual regions.

The distance head applies global average pooling and a Softplus output. Let $d_i^* = d_i/500$ be the normalized reference distance and $e_i = \hat{d}_i - d_i^*$; we use the Huber loss:

$$\mathcal{L}_{\text{d}} = \frac{1}{N} \sum_{i=1}^N \ell_{\text{H}}(e_i), \quad \ell_{\text{H}}(e) = \begin{cases} \frac{1}{2} e^2, & |e| < 1, \\ |e| - \frac{1}{2}, & |e| \geq 1, \end{cases} \quad (4)$$

where $\hat{d}_i$ is the predicted normalized distance.

The complete training objective is

$$\mathcal{L} = \lambda_{\text{det}} \mathcal{L}_{\text{det}} + \mathcal{L}_{\text{r}} + \mathcal{L}_{\text{g}} + \lambda_{\text{d}} \mathcal{L}_{\text{d}}. \quad (5)$$

2.3. Audio-only inference

For Track A, the nine-band acoustic map is used as the only input. The final audio-only system uses local RoI masks as the primary output, while the full-frame decoder mainly acts as a training-time regularizer. During inference, detections are score-filtered, processed by per-class NMS, temporally linked by category-constrained IoU matching, and exported in the official JSON format.

2.4. Audio-visual extension

For Track B, RGB and acoustic inputs are first processed by separate feature extractors. The visual stream uses a pretrained ResNet-50-FPN, while the acoustic stream maps UpLAM features into a lightweight pyramid aligned with the visual FPN levels. As shown in Fig. 1, cross-modal FPN fusion injects acoustic spatial evidence into visual features through gated residual connections. The residual design keeps the pretrained visual representation stable and lets the acoustic branch act as a localization guide rather than an unconstrained replacement for visual features.

A compact ResNet-Conformer context adapter is inserted into the feature pipeline to combine local convolutional modeling with self-attention over panoramic tokens [21], [22]. After proposal generation, we apply RoI-level cross-modal acoustic-visual fusion (RoI-CMAF). For each proposal, visual and acoustic RoI descriptors are fused by attention and a gated residual update. The fused descriptor is mainly used by the classification and energy-mask branches, while localization remains visual-dominant. This separation reduces unstable box updates caused by diffuse acoustic activations.

We additionally use a pretrained DCASE 2023 AV-SELD model as an inference-time semantic prior [9], [23]. It is used only to refine category labels, count constraints, and ranking scores. The spatial mask itself is still produced by the proposed instance model. Finally, sparse high-energy pixels are converted into compact exported regions using a region-aware hull strategy, improving agreement with the official mask rendering procedure.

3. EXPERIMENTS

3.1. Dataset

We use the STAIRS26 dataset released for DCASE 2026 Task 3. It is derived from STARSS23 real indoor audio-visual recordings [24] and provides the original 32-channel Eigenmike signals, synchronized 360° video, and high-resolution SAI annotations. Unlike conventional SELD datasets that use point-wise DoA trajectories, STAIRS26 represents each active sound-producing instance with a semantic polygon mask that encodes class, spatial extent, position, distance, and normalized acoustic-energy intensity. The development set contains about 7.5 hours over 168 clips, and the official evaluation set contains 79 clips. We use the official four-channel spatial audio for both tracks and the synchronized equirectangular video for Track B.

Annotations are provided at 10 frames per second in JSON format. Each entry includes frame index, instance ID, class ID, source distance,

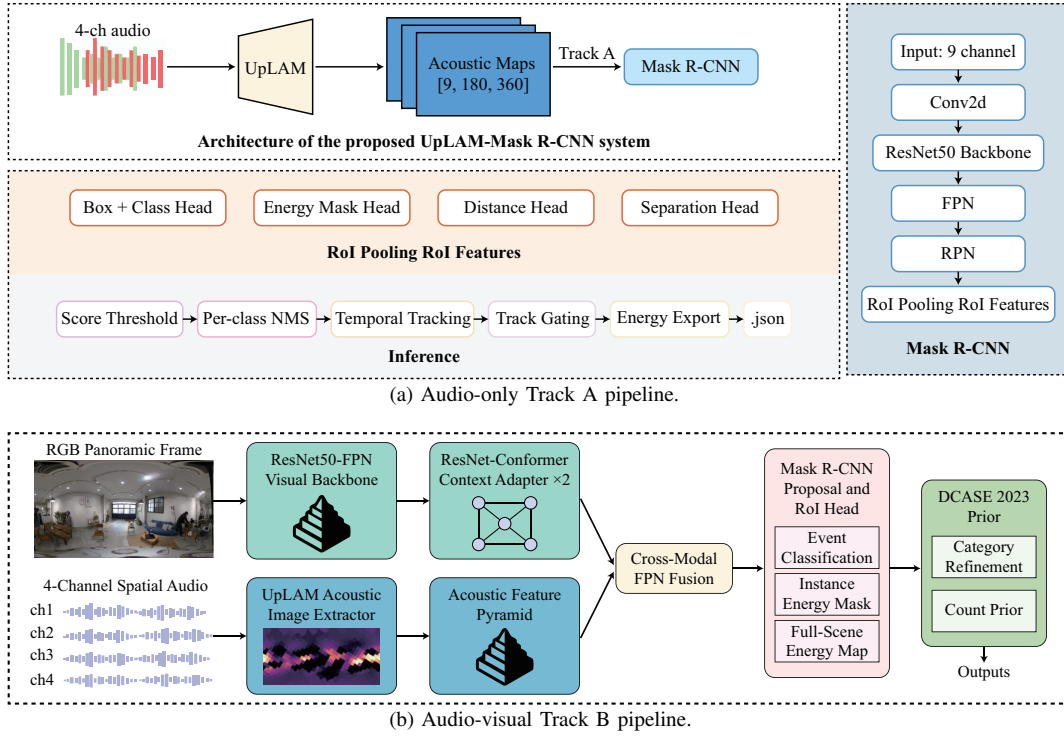


Fig. 1: Overview of the proposed framework. (a) Audio-only Track A predicts SAI instances from UpLAM acoustic maps. (b) Audio-visual Track B adds RGB features, acoustic-guided fusion, RoI-level fusion, and teacher-based semantic priors.

and one or more polygons on a 360×180 ERP grid. Multiple polygons may represent an instance crossing the horizontal wrap-around boundary. The 13 semantic classes follow STARSS23 and DCASE 2026 Task 3 [8], [24]. Training uses audio channel swapping for the audio modality and video perspective shifting for RGB frames, with labels transformed consistently; validation and evaluation clips are left unchanged.

3.2. Setup and metrics

We evaluate on DCASE 2026 Task 3, which uses STARSS23 audio-visual recordings [24] and STAIRS26 high-resolution SAI annotations [8]. Development-set experiments are used to compare variants, while the official evaluation set is used only for the final challenge ranking. All predictions are exported at a temporal resolution of 100 ms, i.e., 10 frames per second, in the official JSON format.

The official evaluation uses two main metrics: Mask mean Average Precision (Mask mAP) and Pearson correlation r . For Mask mAP, predicted and reference SAI instances of the same class are matched according to the IoU between their rendered binary masks. Following the COCO-style protocol [25], AP is averaged over the 13 target classes and ten IoU thresholds as

$$\text{Mask mAP} = \frac{1}{C|\mathcal{T}|} \sum_{c=1}^C \sum_{t \in \mathcal{T}} \text{AP}_c^{(t)}, \quad (6)$$

where $C = 13$ and $\mathcal{T} = \{0.50, 0.55, \dots, 0.95\}$. Macro scores are averaged over classes, while micro scores are computed by pooling all instances before constructing the precision–recall curve.

Pearson correlation r measures the similarity between matched predicted and reference acoustic-energy maps. For a matched pair, r

is computed over the dilated union region $\mathcal{U}$ of the two masks as

$$r = \frac{\sum_{(u,v) \in \mathcal{U}} (E(u,v) - \bar{E}_{\mathcal{U}}) (\hat{E}(u,v) - \bar{\hat{E}}_{\mathcal{U}})}{\sqrt{\sum_{(u,v) \in \mathcal{U}} (E(u,v) - \bar{E}_{\mathcal{U}})^2 \sum_{(u,v) \in \mathcal{U}} (\hat{E}(u,v) - \bar{\hat{E}}_{\mathcal{U}})^2}}. \quad (7)$$

where E and $\hat{E}$ denote the reference and predicted rendered acoustic-energy maps, respectively. The official ranking is based on the cumulative rank of Mask mAP and Pearson r , with Mask mAP used as the tie breaker.

3.3. Implementation details

Both tracks are optimized with AdamW [26] using a base learning rate of 3×10^{-4} , a weight decay of 10^{-4} , and automatic mixed precision. Gradients are clipped to a maximum norm of 2.0. When the smoothed validation loss plateaus, a reduce-on-plateau scheduler reduces the learning rate by a factor of 0.5, with a minimum learning rate of 10^{-7} . We use module-wise learning-rate multipliers: newly added RoI classification/regression heads, RoI-CMAF, ResNet-Conformer, energy-prediction heads, and the distance head use larger learning-rate multipliers, while the pretrained backbone, FPN, and RPN use smaller learning rates or are kept frozen depending on the training stage.

The objective combines detection losses, RoI-level energy reconstruction, full-frame energy supervision, and distance regression. The detection losses include objectness, box regression, and category prediction. The RoI energy head predicts 28×28 instance energy masks and is supervised with weighted MSE, soft-IoU, and Lovasz–Jaccard losses. The full-frame branch predicts a 180×360 scene-level acoustic energy map. The distance branch is trained in earlier stages, but in the final main fine-tuning configuration its loss weight is set to zero; therefore, it is retained as an auxiliary component rather than a dominant optimization target.

Table 1: Track A development-set results.

System	Macro				Micro	
	mAP	AP25	AP50	AP75	mAP	AP50
baseline	0.0003	0.0010	0.0000	0.0000	0.0011	0.0000
FAO	0.0324	0.0721	0.0248	0.0004	0.0222	0.0190
OR-AO	0.0286	0.0598	0.0255	0.0006	0.0220	0.0202

Table 2: Track B development-set results of clean variants.

System	Macro mAP	Macro r	Micro mAP	Micro r
Baseline	0.0003	0.1358	0.0011	0.4429
TPAV	0.0789	0.5366	0.0741	0.5828
ERAV	0.0718	0.5315	0.0709	0.5826
CFAV	0.1508	0.5948	0.0417	0.5948

4. RESULTS AND DISCUSSION

4.1. Development-set analysis

Table 1 reports Track A development-set results. Compared with the official baseline, the final audio-only system (FAO) improves Macro mAP from 0.0003 to 0.0324 and Micro mAP from 0.0011 to 0.0222. However, its AP75 remains low (0.0004), suggesting that fine-grained source-region reconstruction remains challenging in the audio-only setting. FAO gives the strongest macro mAP among our audio-only variants. Objectness-ranked AO (OR-AO) reduces the number of detections and improves AP50, but the drop in macro mAP indicates that aggressive score calibration removes useful detections under easier localization thresholds.

Table 2 summarizes TPAV, ERAV, and CFAV, three Track B variants obtained with different fine-tuning strategies; CFAV additionally uses weak-class fine-tuning. The baseline uses early fusion by concatenating RGB frames and acoustic maps, whereas our systems use separate modality-specific pyramids followed by structured fusion. TPAV gives the best micro mAP, while CFAV achieves the highest macro metrics after weak-class fine-tuning but lower micro mAP. The results show that performance depends on both fusion design and category/source-count calibration.

Table 3 reports a cumulative ablation study on the Track B development set. Starting from the original early-fusion baseline, we progressively add the revised post-processing protocol, the structured audio-visual model, the teacher-based class prior, and the teacher-based count prior.

The ablation results show that the gain comes from multiple stages. Revised post-processing largely improves the original baseline. Under the same post-processing protocol, the structured audio-visual model gives further gains, indicating that the improvement is not solely due to post-processing. Adding class and count priors further improves mAP, showing that teacher-guided semantic and source-count information is useful for suppressing redundant detections and improving instance-level ranking.

Figure 2 gives a qualitative Track B example. Compared with the baseline, the proposed system produces a single compact energy region near the target source and fewer remote false-positive responses. This behavior is consistent with the gains in Macro mAP shown in Table 2.

4.2. Official evaluation results

Table 4 reports the official evaluation-set results from the DCASE 2026 Task 3 ranking page [27]. The Track A submission obtained third place in the team ranking. In Track B, our team obtained first place. We report both the first-ranked Track B system and an additional Pearson-oriented Track B submission because they reflect different trade-offs between spatial precision and energy-map correlation.

Table 3: Cumulative ablation on the Track B development set. The symbol • indicates that the component is enabled, while “–” indicates that it is not used.

Variant	Components				mAP	
	Post.	Struct.	Cls.	Count	Macro	Micro
Original baseline	–	–	–	–	0.0003	0.0011
PP-Baseline	•	–	–	–	0.0222	0.0268
Structured AV	•	•	–	–	0.0258	0.0299
+ Class prior	•	•	•	–	0.0302	0.0364
+ Count prior	•	•	•	•	0.0493	0.0681

Post. denotes the revised post-processing protocol, Struct. denotes the structured audio-visual model, Cls. denotes the teacher-based class prior, and Count denotes the teacher-based count prior.

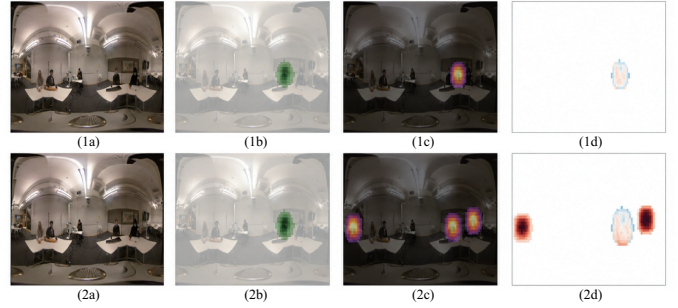

Fig. 2: Qualitative Track B comparison. Top row: proposed system. Bottom row: baseline. Columns show RGB frame, ground truth, prediction, and residual, respectively.

Table 4: Official evaluation-set results of selected submitted systems.

Track	Submission	Rank	mAP	AP50	r
A	Wu_WHU_task3a_1	3 [†]	0.0091	0.0072	0.3185
B	Yang_WHUiasp_task3b_2	1	0.0131	0.0120	0.4002
B	Yang_WHUiasp_task3b_3	3	0.0054	0.0056	0.5634

[†]Track A rank is the official team rank; the table reports the metrics of the corresponding submitted system.

4.3. Discussion

The development and evaluation results reveal a trade-off between detection-oriented mAP and energy-field correlation. In Track B, the first-ranked submission achieves a higher Macro mAP, whereas the Pearson-oriented submission achieves a higher Pearson correlation. This suggests that compact high-IoU instance masks and globally consistent acoustic-energy reconstruction are not always optimized by the same calibration strategy.

The results also support partially separated acoustic and visual feature extraction. UpLAM provides dense spatial cues but may not preserve all spectral-temporal information required for semantic event classification. This may partly explain the remaining gap in Track A, whereas Track B additionally benefits from visual cues and teacher-guided semantic and source-count priors.

5. CONCLUSION

We presented a unified acoustic map-based instance modeling framework for semantic acoustic imaging in DCASE 2026 Task 3. The method represents four-channel spatial audio as UpLAM acoustic maps and predicts event instances with Mask R-CNN-style RoI heads. Track A demonstrates that instance-level SAI prediction can be performed from audio maps alone, while Track B improves audio-visual inference through modality-specific pyramids, gated acoustic guidance, RoI-level fusion, and semantic priors. The official evaluation results show competitive performance in both tracks, including third place in Track A team ranking and first place in Track B.

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